

Software Systems

Assignment 2

System Description: Smart Campus Navigation App

A university is developing a Smart Campus Navigation App to help students and visitors find buildings, classrooms, offices, and campus facilities. The app displays routes, estimated walking times, and accessibility options (e.g., elevators, ramps). Users can search for a faculty member or event location and get step-by-step directions. The system relies on interactions between the User, the Mobile App UI, a backend Navigation Service, and a Campus Map Database.

Question 1: Activity Diagram Task

For the Smart Campus Navigation App, choose two use cases and create Activity Diagrams for them.

Tasks:

Draw Activity Diagrams for any two of the following (or similar) use cases:

- a. Search for a Building
- b. Find Faculty Office
- c. Get Directions
- d. Search for Event Location
- e. View Accessibility Options

Each Activity Diagram must include:

- a. Initial node
- b. Activities/actions
- c. Decision nodes / guards
- d. Merge nodes
- e. Swimlanes (if applicable) to separate responsibilities (e.g., User, Mobile App, Navigation Service)
- f. Final node

Question 2: Sequence Diagram Task

Model the dynamic interaction between the system components over time for a specific scenario.

Tasks:

Draw a Sequence Diagram illustrating the execution of the "Search for a Faculty Office and Get Directions" process. Your Sequence Diagram must clearly include:

- **Lifelines / Objects:** Include the actor (User) and at least three internal system objects (e.g., Mobile App UI, Navigation Service, Map Database).
- **Messages:** Accurately use UML notation for synchronous calls (solid arrow with filled head), asynchronous calls (solid arrow with open head), and return messages (dashed arrow).
- **Focus of Control:** Use activation bars (rectangles on lifelines) to depict when an object is actively executing a process.
- **Combined Fragments:** Include at least one **alt** (alternative) block to handle the scenario where the faculty office is successfully found in the database versus when it is not found.

Submission Guidelines

- Submit as a single .pdf or .docx file.
- Include two Activity Diagrams and one Sequence Diagram.
- Use clear UML notation (Lucidchart, Draw.io, or Visual Paradigm recommended).
- Label all messages, states, activities, decisions, and swimlanes properly. Ensure diagrams are readable and consistently formatted.